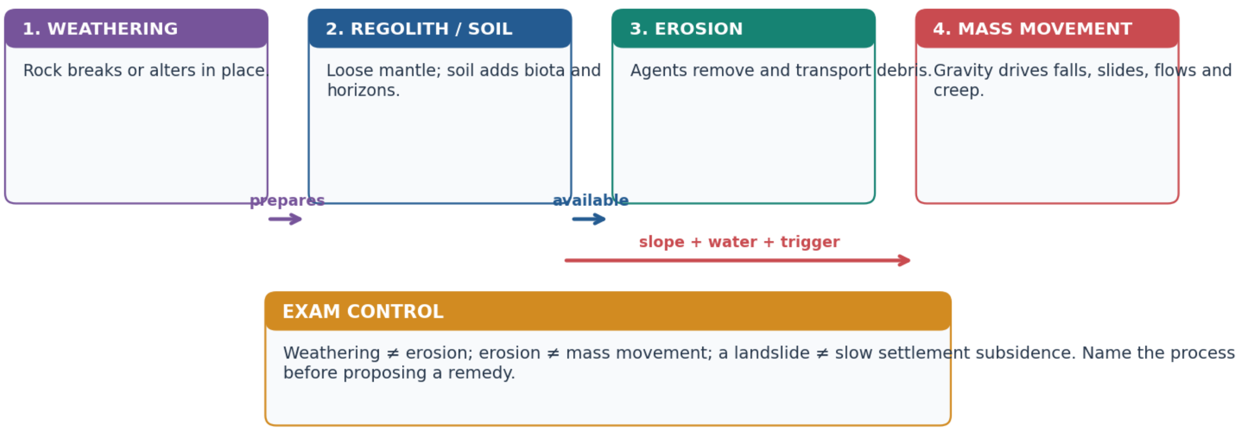
SOLVED PRACTICE WORKBOOK

Geography 04 - Weathering, Mass Movement, Groundwater / India Erosion-Landslides-Groundwater - Solved Practice Workbook

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

Weathering → regolith → erosion / mass movement: one linked system

Breakdown is in place; removal needs an agent; gravity can move material without a transport agent.



Original deterministic schematic • not to scale • conceptual learning aid

Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

CONTENTS / WORKBOOK INDEX

Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

Weathering, Mass Movement, Groundwater / India Erosion-Landslides-Groundwater - Solved Practice Workbook

BASIC MCQS / REMEDIATION

Practice rule

These 48 questions are original UPSC-style practice. The generation pipeline uses the stable `geography-04` seed to balance and non-pattern correct-option positions while preserving the correct answer content. Questions 41-48 are remedial.

MCQ 1 - Exogenic boundary

Which statement correctly separates weathering from erosion?

A. Weathering changes rock in situ, whereas erosion includes detachment and transport by an external agent. B. Weathering always moves sediment downhill. C. Erosion occurs only after soil has formed. D. Denudation excludes mass wasting.

Answer: A.

Explanation: Weathering is defined by absence of transport; denudation is the wider lowering system.

MCQ 2 - Denudation system

Which sequence best represents a source-transfer-storage system?

A. Deposition -> uplift -> no further change B. Weathering -> regolith -> entrainment/transport -> deposition -> possible remobilisation C. Mass wasting -> weathering only after transport D. Soil formation -> no erosion

Answer: B.

Explanation: Weathered material can move, be stored and later be remobilised.

MCQ 3 - Weathering controls

Which is the most accurate statement?

A. Climate alone fixes the same weathering product in every rock. B. Rock structure matters only in deserts. C. Climate interacts with mineralogy, joints, relief, organisms, time and land use to control weathering. D. Time always creates deep soil even on rapidly eroding slopes.

Answer: C.

Explanation: Weathering is a multivariable process.

MCQ 4 - Thermal and unloading

Which distinction is correct?

A. Both terms mean frost action. B. Thermal expansion changes feldspar directly to clay. C. Unloading requires salt crystallisation. D. Thermal expansion stresses rock through temperature cycling; unloading creates pressure-release sheet joints.

Answer: D.

Explanation: The two mechanisms are mechanical but not identical.

MCQ 5 - Frost action

Frost weathering is strongest where:

A. Water repeatedly enters joints, freezes and thaws around the freezing point. B. The rock remains permanently dry. C. Temperatures never approach freezing. D. Only chemical solution occurs.

Answer: A.

Explanation: Water access and repeated freeze-thaw cycles are essential.

MCQ 6 - Salt crystallisation

Salt weathering most directly involves:

A. A falling water table around a pumping well. B. Crystal growth or hydration pressure within pores and cracks. C. Downslope sliding along a bedding plane. D. Carbonic acid dissolving limestone only.

Answer: B.

Explanation: Evaporation concentrates salts and growing crystals disrupt rock.

MCQ 7 - Wetting and drying

Repeated wetting and drying especially weakens:

A. Only unjointed quartzite. B. A confined aquifer by artesian flow. C. Clay-rich material that swells and shrinks. D. Freshwater head at a coast only.

Answer: C.

Explanation: Moisture cycling creates fatigue in expansive materials.

MCQ 8 - Carbonation

Carbonation is best described as:

A. Oxygen reacting with iron only. B. Mechanical removal by wind. C. Physical expansion during freezing. D. Carbon-dioxide-bearing water forming weak carbonic acid that dissolves carbonate minerals.

Answer: D.

Explanation: Carbonation is a chemical dissolution pathway central to limestone.

MCQ 9 - Hydrolysis

Hydrolysis can transform feldspar mainly into:

A. Clay minerals and dissolved ions. B. Fresh volcanic glass without alteration. C. Pure metallic iron. D. Glacial till.

Answer: A.

Explanation: Water reacts with silicate minerals and creates secondary minerals.

MCQ 10 - Oxidation-reduction

Which statement is correct?

A. Oxidation requires no oxygen-bearing pathway. B. Iron-bearing minerals may oxidise in oxygenated conditions, while reducing conditions can change iron to more soluble forms. C. Reduction is a mechanical fracture process. D. Redox has no relation to drainage.

Answer: B.

Explanation: Redox state depends strongly on oxygen and water conditions.

MCQ 11 - Biological weathering

Plant roots and microbes can:

A. Operate only after erosion has ended. B. Create artesian pressure. C. Widen joints physically and accelerate chemical alteration through organic acids. D. Prevent all weathering beneath vegetation.

Answer: C.

Explanation: Biological action commonly reinforces physical and chemical processes.

MCQ 12 - Regolith and soil

Which statement is correct?

A. Soil forms without parent material, organisms or time. B. All regolith is mature fertile soil. C. The B horizon is always unweathered bedrock. D. Regolith is weathered mantle; soil requires biological activity and horizon development.

Answer: D.

Explanation: Soil is an organised natural body, not merely broken rock.

MCQ 13 - Eluviation and illuviation

Which pairing is correct?

A. Eluviation is removal from an upper horizon; illuviation is accumulation in a lower horizon. B. Both mean surface erosion. C. Illuviation occurs only in deserts. D. Eluviation means artesian flow.

Answer: A.

Explanation: The pair describes vertical translocation within a soil profile.

MCQ 14 - Black soil

Black cotton soil is most closely linked to:

A. Recent marine coral deposits. B. Weathering of Deccan basaltic/fissure-volcanic parent material and clay-rich shrink-swell behaviour. C. Granite weathering alone in a perhumid climate. D. Unweathered shale with no clay.

Answer: B.

Explanation: Parent material and seasonal moisture help explain regur properties.

MCQ 15 - Sheet-rill-gully

Which order shows increasing concentration and incision by runoff?

A. Ravine -> sheet -> rill -> gully B. Gully -> sheet -> ravine -> rill C. Sheet -> rill -> gully -> ravine D. Rill -> ravine -> sheet -> gully

Answer: C.

Explanation: Runoff becomes progressively channelised and erosive.

MCQ 16 - Wind erosion

Which transport sequence is associated with wind erosion?

A. Solution, carbonation and hydrolysis B. Infiltration, percolation and recharge C. Creep, slump and artesian rise D. Surface creep, saltation and suspension

Answer: D.

Explanation: Wind moves coarse grains near the surface and finer grains in suspension.

MCQ 17 - Creep

A classic field indicator of soil creep is:

A. Slowly tilted fences, posts or tree trunks on a slope. B. A flowing artesian well. C. A single deep ocean trench. D. A fresh lava flow.

Answer: A.

Explanation: Creep is gradual and produces subtle deformation.

MCQ 18 - Solifluction

Solifluction refers to:

A. Dry rockfall from a vertical cliff only. B. Slow flow of water-saturated surface material over frozen or impermeable ground. C. Chemical oxidation of iron. D. Coastal saline upconing.

Answer: B.

Explanation: Saturation above a restricting layer promotes slow lobate movement.

MCQ 19 - Slump

A slump is distinguished by:

A. Only wind transport. B. Free fall without a failure surface. C. Rotational movement along a curved failure surface, often with back-tilted blocks. D. Uniform sheet wash.

Answer: C.

Explanation: Rotational geometry is the diagnostic clue.

MCQ 20 - Debris flow

Which description best fits a debris flow?

A. A coherent block moving on one plane only. B. Imperceptible soil creep. C. Groundwater moving through an aquitard. D. A rapid, water-rich, deforming mixture of coarse and fine material, often channelised.

Answer: D.

Explanation: Water and mixed debris produce flow-like motion.

MCQ 21 - Slope threshold

A slope becomes unstable when:

A. Driving shear stress exceeds available resisting shear strength. B. Porosity becomes zero everywhere. C. Rainfall begins regardless of material and slope. D. An inventory map is published.

Answer: A.

Explanation: Failure is a stress-strength threshold problem.

MCQ 22 - Pore-water pressure

Why can saturation reduce slope stability?

A. It always cements grains permanently. B. It may add weight and raise pore-water pressure, lowering effective resistance. C. It removes all joints. D. It converts every slide into subsidence.

Answer: B.

Explanation: Water can increase driving stress while reducing effective friction.

MCQ 23 - Toe undercutting

Toe erosion or excavation can destabilise a slope because it:

A. Always increases cohesion. B. Raises the potentiometric surface in all aquifers. C. Removes support from the lower slope and may steepen the profile. D. Prevents runoff concentration.

Answer: C.

Explanation: Loss of toe support changes slope-force balance.

MCQ 24 - Himalaya-Ghats comparison

Which comparison is most defensible?

A. Himalayan landslides have no rainfall link. B. Both regions have identical tectonic history. C. Western Ghats landslides are always earthquake-triggered. D. The Himalaya combines young tectonic relief and fractured material; the Western Ghats often combine weathered escarpments, intense orographic rain and land-use/drainage change.

Answer: D.

Explanation: The regions share triggers but differ in geological predisposition.

MCQ 25 - Inventory versus forecast

A landslide inventory primarily records:

A. Past mapped or reported events and their attributes. B. Only rainfall totals. C. The exact date of every future failure. D. Groundwater quality samples.

Answer: A.

Explanation: Inventory supports pattern analysis but is not deterministic prediction.

MCQ 26 - Susceptibility map

A susceptibility map most directly shows:

A. Guaranteed annual landslide counts. B. Relative spatial propensity based on terrain and conditioning factors. C. Only exposed population. D. The official compensation amount.

Answer: B.

Explanation: Susceptibility is a spatial screening product.

MCQ 27 - Early warning

Which statement about landslide early warning is correct?

A. It removes weak rock. B. It predicts every failure with certainty. C. It can reduce exposure if thresholds, communication and action work, but it does not stabilise the slope. D. It replaces drainage and land-use planning.

Answer: C.

Explanation: Warning and source mitigation perform different functions.

MCQ 28 - Subsidence versus landslide

Which distinction is correct?

A. Both are always identical. B. A landslide has no shear movement. C. Subsidence can occur only at coasts. D. Subsidence is ground lowering/settlement; a landslide is downslope mass movement.

Answer: D.

Explanation: They can coexist but should not be used as synonyms.

MCQ 29 - Joshimath evidence

The safest Joshimath conclusion is:

A. Use a multi-causal frame involving ground material, drainage/water, slope and loading, with integrated evidence. B. All change is caused by groundwater pumping alone. C. One visible crack proves one cause. D. Monitoring removes uncertainty.

Answer: A.

Explanation: The case requires bounded, evidence-led attribution.

MCQ 30 - Water budget

Which equation is conceptually correct?

A. Storage always equals rainfall. B. Change in groundwater storage = recharge + inflow - discharge - pumping - outflow. C. Discharge is not part of the aquifer budget. D. Pumping creates recharge automatically.

Answer: B.

Explanation: Groundwater storage changes through multiple inflows and outflows.

MCQ 31 - Infiltration-percolation

Which distinction is correct?

A. Both mean pumping. B. Percolation happens before surface entry. C. Infiltration is entry at the surface; percolation is subsequent downward movement through material. D. Infiltration always reaches the aquifer.

Answer: C.

Explanation: Surface entry does not guarantee deep recharge.

MCQ 32 - Porosity-permeability

Why can clay be porous but poorly permeable?

A. It contains no water. B. It is always crystalline limestone. C. Permeability means total pore volume only. D. It may contain many tiny pores that are poorly connected for rapid flow.

Answer: D.

Explanation: Storage space and connected transmission are different properties.

MCQ 33 - Aquifer-aquitard

Which pairing is correct?

A. Aquifer-stores and transmits usable groundwater; aquitard-transmits much more slowly. B. Aquifer-zone of aeration only; aquitard-water table. C. Aquifer-surface river; aquitard-rainfall. D. Aquifer-completely impermeable; aquitard-unlimited flow.

Answer: A.

Explanation: The distinction is relative hydraulic transmission.

MCQ 34 - Confined aquifer

A confined aquifer is:

A. Always exposed directly at the local ground surface. B. A permeable unit bounded by lower-permeability layers and commonly under pressure. C. Necessarily saline. D. Unable to recharge anywhere.

Answer: B.

Explanation: Confinement changes head conditions but does not create unlimited water.

MCQ 35 - Artesian flow

A well flows at the surface without pumping only when:

A. The aquifer is unconfined and dry. B. The well is merely deep. C. The potentiometric level of the confined aquifer lies above ground level. D. Clay has high permeability.

Answer: C.

Explanation: Artesian rise is pressure-controlled and flowing conditions are conditional.

MCQ 36 - Perched aquifer

A perched water body forms when:

A. The entire basin becomes confined. B. Seawater rises under every coastal well. C. A river erodes a ravine. D. A local low-permeability lens holds saturation above the regional water table.

Answer: D.

Explanation: Perched aquifers are local and commonly seasonal.

MCQ 37 - Hard-rock aquifer

Groundwater in crystalline hard-rock terrain is commonly stored and transmitted through:

A. Weathered mantles, joints, fractures and other secondary openings. B. Only primary intergranular pores like loose sand. C. An unlimited underground lake. D. Clay aquicludes alone.

Answer: A.

Explanation: Secondary porosity creates highly variable local yields.

MCQ 38 - Cone of depression

A cone of depression is:

A. A gully head in a ravine. B. The local lowering of hydraulic head around a pumping well. C. The saline interface at the coast only. D. A soil horizon.

Answer: B.

Explanation: Pumping produces spatial drawdown around the well.

MCQ 39 - Safe and sustainable yield

Which statement is most accurate?

A. Sustainable yield ignores baseflow and springs. B. Safe yield always equals annual recharge. C. Sustainable yield considers ecological, quality, social and long-term effects beyond a simple recharge-volume rule. D. Both terms mean unlimited pumping.

Answer: C.

Explanation: Groundwater sustainability is broader than arithmetic balance.

MCQ 40 - Coastal intrusion

Excessive coastal pumping can:

A. Push the saline interface seaward automatically. B. Remove all salinity from soil. C. Prove surface storm-surge flooding. D. Lower freshwater head and induce landward interface movement or local upconing.

Answer: D.

Explanation: Hydraulic-head decline is the central mechanism.

MCQ 41 - Remedial: weathering versus erosion

A rock cracks in place and is later carried away by a stream. Which classification is correct?

A. Cracking is weathering; stream removal is erosion and transportation. B. Both stages are deposition. C. Cracking is erosion because rock changed. D. Stream transport is weathering.

Answer: A.

Explanation: Separate in-situ breakdown from agent-driven removal.

MCQ 42 - Remedial: infiltration versus percolation

Rain enters the soil but is later stopped by a compact layer. What is certain?

A. Recharge must equal rainfall. B. Infiltration occurred, but deep percolation/recharge may have been limited. C. No infiltration occurred. D. The aquifer became artesian.

Answer: B.

Explanation: Entry and deep downward transfer are different.

MCQ 43 - Remedial: porosity versus permeability

Which material can have high porosity but low permeability?

A. Open well-sorted gravel only. B. A fully open fracture. C. Clay with many small poorly connected pores. D. A river channel.

Answer: C.

Explanation: Clay is the standard close-option trap.

MCQ 44 - Remedial: aquifer type

Water rises in a well above the top of the tapped aquifer but not above ground. This indicates:

A. No pressure. B. An unconfined water table at ground level. C. A perched aquifer necessarily. D. A confined artesian condition that is not flowing at the surface.

Answer: D.

Explanation: Artesian means pressure rise; flowing artesian is a narrower case.

MCQ 45 - Remedial: landslide type

A coherent mass moves downslope along a curved surface and leaves a head scarp. It is most likely:

A. A rotational slide or slump. B. A rockfall. C. Soil creep only. D. Saltation.

Answer: A.

Explanation: Curved failure geometry and back rotation distinguish a slump.

MCQ 46 - Remedial: rainfall causation

After intense rain a slope fails. Which conclusion is valid?

A. Rain is necessarily the sole cause. B. Rain may be the trigger, while material, structure, drainage and cutting require investigation as predispositions. C. Road cutting can never matter. D. The event was predicted by any susceptibility map.

Answer: B.

Explanation: Trigger and predisposition must be separated.

MCQ 47 - Remedial: quantity-quality

A falling water table proves:

A. Arsenic contamination everywhere. B. Seawater intrusion in every inland basin. C. Only a quantity trend at the measured location; contamination requires separate sampling and pathway evidence. D. That recharge structures are effective.

Answer: C.

Explanation: Quantity and quality evidence are distinct.

MCQ 48 - Remedial: recharge effectiveness

Which is the best evaluation of a recharge programme?

A. Count structures only. B. Assume every borewell is a recharge shaft. C. Ignore extraction changes. D. Check aquifer fit, source-water quality, maintenance, demand and measured level/quality trends.

Answer: D.

Explanation: Inputs must be separated from hydrological outcomes.

PYQS AND ANSWER PRACTICE

PYQ control note

- Exact wording below was checked against the routed local official papers.
- **Official Set-A keys:** 2024 Q7 = **A**; 2025 Q28 = **C**.
- **Key unavailable locally:** 2020 Q98, 2021 Q64 and 2023 Q76. Any independent solution is explicitly non-official.
- Mains has no UPSC model key; the answers below are independent examiner-grade models.
- Direct owner is shown first. Cross-owned questions are included because Topic 04 supplies essential physical-geography mechanics, but ownership is not transferred.

VERIFIED PYQ 1 - UPSC Prelims 2021 GS-I Q64: black cotton soil

Exact question: The black cotton soil of India has been formed due to the weathering of

A. brown forest soil B. fissure volcanic rock C. granite and schist D. shale and limestone

Official-key status: No official local key is available.

Independent solution - NOT AN OFFICIAL KEY: B. The Deccan black-soil region is associated with weathering of basaltic lava flows, described in the option as fissure volcanic rock. Brown forest soil is not parent rock; granite/schist and shale/limestone do not explain the classic regur association.

VERIFIED PYQ 2 - UPSC Prelims 2023 GS-I Q76: groundwater withdrawal

Exact question:

Statement-I: According to the United Nations ‘World Water Development Report, 2022', India extracts more than a quarter of the world's groundwater withdrawal each year.

Statement-II: India needs to extract more than a quarter of the world's groundwater each year to satisfy the drinking water and sanitation needs of almost 18% of world's population living in its territory.

Which one of the following is correct in respect of the above statements?

A. Both Statement-I and Statement-II are correct and Statement-II is the correct explanation for Statement-I B. Both Statement-I and Statement-II are correct and Statement-II is not the correct explanation for Statement-I C. Statement-I is correct but Statement-II is incorrect D. Statement-I is incorrect but Statement-II is correct

Official-key status: No official local key is available.

Independent solution - NOT AN OFFICIAL KEY: C. Statement-I is consistent with the cited report framing. Statement-II is not a valid explanation because India's groundwater withdrawal is dominated by irrigation rather than a necessary drinking-water-and-sanitation requirement proportional to population. The 10 August 2026 Ministry reply separately reports 87% irrigation, 11% domestic and 2% industrial shares for the 2025 assessment.

VERIFIED PYQ 3 - UPSC Prelims 2024 GS-I Q7: rainfall and weathering

Exact question:

Statement-I: Rainfall is one of the reasons for weathering of rocks. Statement-II: Rain water contains carbon dioxide in solution. Statement-III: Rain water contains atmospheric oxygen.

Which one of the following is correct in respect of the above statements?

A. Both Statement-II and Statement-III are correct and both of them explain Statement-I B. Both Statement-II and Statement-III are correct, but only one of them explains Statement-I C. Only one of the Statements II and III is correct and that explains Statement-I D. Neither Statement-II nor Statement-III is correct

Official local Set-A key: A. Dissolved carbon dioxide enables weak carbonic-acid/carbonation reactions; dissolved oxygen supports oxidation. Both can make rainfall a weathering agent.

VERIFIED PYQ 4 - UPSC Prelims 2025 GS-I Q28: chalk and clay

Exact question:

Statement-I: In the context of effect of water on rocks, chalk is known as a very permeable rock whereas clay is known as quite an impermeable or least permeable rock. Statement-II: Chalk is porous and hence can absorb water. Statement-III: Clay is not at all porous.

Which one of the following is correct in respect of the above statements?

A. Both Statement-II and Statement-III are correct and both of them explain Statement-I B. Both Statement-II and Statement-III are correct but only one of them explains Statement-I C. Only one of the Statements II and III is correct and that explains Statement-I D. Neither Statement-II nor Statement-III is correct

Official local Set-A key: C. Statement-II is correct and explains chalk's ability to admit water; Statement-III is false because clay can be porous but has very low permeability due to tiny, poorly connected pores.

SUPPLEMENTARY CROSS-OWNED PYQ 5 - UPSC Prelims 2020 GS-I Q98: CGWA

Exact question:

Consider the following statements:

1. 36% of India's districts are classified as "overexploited" or "critical" by the Central Ground Water Authority (CGWA).
2. CGWA was formed under the Environment (Protection) Act.
3. India has the largest area under groundwater irrigation in the world.

Which of the statements given above is/are correct?

A. 1 only B. 2 and 3 only C. 2 only D. 1 and 3 only

Official-key status: No official local key is available. The workbook does not convert an external or inferred key into an official fact. The concept route is institutional: CGWA was constituted under the Environment (Protection) Act framework; dated percentages must be treated cautiously.

VERIFIED PYQ 6 - UPSC Mains 2018 GS-I Q14, 15 marks

Exact question: "The ideal solution of depleting ground water resources in India is water harvesting system." How can it be made effective in urban areas? (Answer in 250 words)

Model answer

Urban water harvesting can relieve groundwater stress only when it is designed as a water-budget and aquifer-management system rather than as a construction target.

First, cities should capture roof runoff for direct non-potable storage and route only suitably treated surplus to recharge. First-flush diversion, filtration and separation from sewage are essential because contaminated recharge can damage an aquifer. Second, recharge structures must follow hydrogeological mapping: permeable recharge zones, depth to water, confining layers and contamination risk decide whether a pit, shaft, trench, well or restored tank is suitable.

Third, urban planning must preserve lakes, wetlands, floodplains and open recharge areas while reducing impervious cover through blue-green infrastructure. Storm-water drains should be integrated with detention and recharge rather than used merely to export runoff. Fourth, demand must fall through leakage control, reuse of treated wastewater, metering/regulation of bulk extraction and water-efficient fixtures; otherwise added recharge is overwhelmed.

Finally, municipal bodies need enforceable building provisions, maintenance registers, public aquifer/water-level maps and quality monitoring. The 3-10 August 2026 Ministry of Jal Shakti replies confirm that CGWB/NAQUIM, digital monitoring and recharge campaigns provide planning instruments, but structure counts do not prove recovery.

Conclusion: harvesting is effective when **capture + quality + aquifer fit + demand control + maintenance + measured outcomes** operate together.

Why this earns marks: It answers "how effective", uses an urban-specific mechanism, names current institutions and adds an outcome-based qualification.

Demand decoding: The directive **answer** requires a direct position on "VERIFIED PYQ 6 - UPSC Mains 2018 GS-I Q14, 15 marks", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in "VERIFIED PYQ 6 - UPSC Mains 2018 GS-I Q14, 15 marks".

Analytical body:

1. **Claim and named evidence:** VERIFIED PYQ 6 - UPSC Mains 2018 GS-I Q14, 15 marks **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** Exact question: "The ideal solution of depleting ground water resources in India is water harvesting system." How can it be made effective in urban areas? (Answer in 250 words) **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** Urban water harvesting can relieve groundwater stress only when it is designed as a water-budget and aquifer-management system rather than as a construction target. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
4. **Claim and named evidence:** Conclusion: harvesting is effective when capture + quality + aquifer fit + demand control + maintenance + measured outcomes operate together. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
5. **Claim and named evidence:** Why this earns marks: It answers "how effective", uses an urban-specific mechanism, names current institutions and adds an outcome-based qualification. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in "VERIFIED PYQ 6 - UPSC Mains 2018 GS-I Q14, 15 marks".

Executable exam-length answer / compression plan: For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

How to improve this answer: For "VERIFIED PYQ 6 - UPSC Mains 2018 GS-I Q14, 15 marks", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

SUPPLEMENTARY CROSS-OWNED PYQ 7 - UPSC Mains GS-III 2019 Q18, 15 marks

Exact question: Disaster preparedness is the first step in any disaster management process. Explain how hazard zonation mapping will help disaster mitigation in the case of landslides. (Answer in 250 words)

Model answer

Landslide hazard zonation maps classify terrain by relative likelihood or intensity of slope failure using geology, slope, drainage, land cover, geomorphology and past-event inventory. They support mitigation before a trigger becomes a disaster.

At planning stage, the maps can keep settlements, hospitals, schools and lifeline corridors away from very-high-susceptibility slopes; guide alignment and design of roads, tunnels and utilities; and identify where site investigation must precede construction. At preparedness stage, zonation helps locate monitoring instruments, rainfall thresholds, evacuation routes, shelters, equipment and maintenance priorities. At mitigation stage, it supports targeted drainage, toe protection, retaining/anchoring, bioengineering and controlled spoil disposal rather than uniform treatment.

GSI's national inventory and susceptibility work provides the spatial base. Its June 2026 Chanmari West assessment illustrates why maps must be followed by field examination: a mapped high-susceptibility slope still required joint, toe-erosion, runoff and settlement-drainage diagnosis.

However, susceptibility is not the same as a date-specific forecast, and hazard is not risk. Exposure, vulnerability, map scale, changing land use, rainfall uncertainty and last-mile communication determine outcomes.

Conclusion: zonation is most effective when legally linked to land-use control, engineering design, maintenance, forecast dissemination and community drills.

Why this earns marks: It explains the decision uses of zonation and qualifies scale, timing and risk limitations.

Demand decoding: The directive **answer** requires a direct position on "SUPPLEMENTARY CROSS-OWNED PYQ 7 - UPSC Mains GS-III 2019 Q18, 15 marks", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in "SUPPLEMENTARY CROSS-OWNED PYQ 7 - UPSC Mains GS-III 2019 Q18, 15 marks".

Analytical body:

1. **Claim and named evidence:** SUPPLEMENTARY CROSS-OWNED PYQ 7 - UPSC Mains GS-III 2019 Q18, 15 marks **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** Exact question: Disaster preparedness is the first step in any disaster management process. Explain how hazard zonation mapping will help disaster mitigation in the case of landslides. (Answer in 250 words) **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** However, susceptibility is not the same as a date-specific forecast, and hazard is not risk. Exposure, vulnerability, map scale, changing land use, rainfall uncertainty and last-mile communication determine outcomes. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

4. **Claim and named evidence:** Conclusion: zonation is most effective when legally linked to land-use control, engineering design, maintenance, forecast dissemination and community drills. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
5. **Claim and named evidence:** Why this earns marks: It explains the decision uses of zonation and qualifies scale, timing and risk limitations. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in "SUPPLEMENTARY CROSS-OWNED PYQ 7 - UPSC Mains GS-III 2019 Q18, 15 marks".

Executable exam-length answer / compression plan: For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

How to improve this answer: For "SUPPLEMENTARY CROSS-OWNED PYQ 7 - UPSC Mains GS-III 2019 Q18, 15 marks", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

SUPPLEMENTARY CROSS-OWNED PYQ 8 - UPSC Mains GS-I 2021 Q4, 10 marks

Exact question: Differentiate the causes of landslides in the Himalayan region and Western Ghats. (Answer in 150 words)

Model answer

Both regions experience gravity-driven failure, but their predispositions differ.

Himalayan region	Western Ghats
Young collision belt, active uplift and seismicity	Older escarpment/plateau-margin terrain
Very high relief, deep river incision and fractured/folded rocks	Deep weathering, lateritic/altered mantle and steep scarps
Snowmelt, monsoon saturation and earthquakes can trigger failure	Intense orographic monsoon rainfall and short saturated slopes dominate many events
Road cutting, tunnelling, toe erosion and unstable debris amplify risk	Quarrying, road/house cuts, plantation/land-use change and blocked drainage amplify risk

In both, rainfall is often a trigger rather than the entire cause; lithology, joints, slope geometry, drainage and human loading need site evidence.

Conclusion: mitigation must therefore be region-specific-tectonic and corridor-sensitive in the Himalaya, and drainage/weathered-mantle/land-use sensitive in the Ghats.

Why this earns marks: It obeys "differentiate", gives paired causal categories and retains a shared multi-factor qualification.

Demand decoding: The directive **answer** requires a direct position on "SUPPLEMENTARY CROSS-OWNED PYQ 8 - UPSC Mains GS-I 2021 Q4, 10 marks", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in "SUPPLEMENTARY CROSS-OWNED PYQ 8 - UPSC Mains GS-I 2021 Q4, 10 marks".

Analytical body:

1. **Claim and named evidence:** SUPPLEMENTARY CROSS-OWNED PYQ 8 - UPSC Mains GS-I 2021 Q4, 10 marks **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** Exact question: Differentiate the causes of landslides in the Himalayan region and Western Ghats. (Answer in 150 words) **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** Both regions experience gravity-driven failure, but their predispositions differ. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
4. **Claim and named evidence:** Young collision belt, active uplift and seismicity Older escarpment/plateau-margin terrain **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
5. **Claim and named evidence:** Very high relief, deep river incision and fractured/folded rocks Deep weathering, lateritic/altered mantle and steep scarps **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in "SUPPLEMENTARY CROSS-OWNED PYQ 8 - UPSC Mains GS-I 2021 Q4, 10 marks".

Executable exam-length answer / compression plan: For a 10-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise three process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

How to improve this answer: For "SUPPLEMENTARY CROSS-OWNED PYQ 8 - UPSC Mains GS-I 2021 Q4, 10 marks", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

SUPPLEMENTARY CROSS-OWNED PYQ 9 - UPSC Mains GS-III 2021 Q18, 15 marks

Exact question: Describe the various causes and the effects of landslides. Mention the important components of the National Landslide Risk Management Strategy. (Answer in 250 words)

Model answer

Landslides are gravity-driven movements of rock, debris or earth when driving stress exceeds resisting strength.

Causes: steep or undercut slopes; weak, weathered or jointed lithology; adverse bedding; old debris; saturation and pore-pressure rise during intense/prolonged rain; river/toe erosion; earthquake shaking; and human triggers such as road cutting, quarrying, loading, leakage, deforestation and unplanned drainage.

Effects: deaths and injuries; burial of settlements; road, rail, pipeline and power disruption; river blockage and flash-flood risk; sedimentation; loss of soil and forests; isolation of mountain communities; and recurring livelihood/fiscal costs.

Risk-management components: inventory and susceptibility/hazard zonation; site investigation and monitoring; rainfall/threshold-based forecasting with last-mile warning; land-use regulation and safer corridor alignment; drainage and slope treatment; bioengineering and designed retaining/toe works; enforcement of hill construction standards; community awareness, evacuation routes and drills; emergency access and post-event learning.

The GSI/NLFC ecosystem supplies inventory, susceptibility and forecast inputs, while NDMA/state authorities own broader disaster-management coordination. Forecasts reduce exposure but do not stabilise a slope.

Conclusion: an effective strategy connects **map -> regulate -> drain/stabilise -> warn -> evacuate -> update inventory**.

Why this earns marks: It covers causes, effects and strategy as three explicit demands and respects institutional ownership.

Demand decoding: The directive **answer** requires a direct position on "SUPPLEMENTARY CROSS-OWNED PYQ 9 - UPSC Mains GS-III 2021 Q18, 15 marks", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in "SUPPLEMENTARY CROSS-OWNED PYQ 9 - UPSC Mains GS-III 2021 Q18, 15 marks".

Analytical body:

1. **Claim and named evidence:** SUPPLEMENTARY CROSS-OWNED PYQ 9 - UPSC Mains GS-III 2021 Q18, 15 marks **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** Exact question: Describe the various causes and the effects of landslides. Mention the important components of the National Landslide Risk Management Strategy. (Answer in 250 words) **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** Landslides are gravity-driven movements of rock, debris or earth when driving stress exceeds resisting strength. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
4. **Claim and named evidence:** The GSI/NLFC ecosystem supplies inventory, susceptibility and forecast inputs, while NDMA/state authorities own broader disaster-management coordination. Forecasts reduce exposure but do not stabilise a slope. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
5. **Claim and named evidence:** Conclusion: an effective strategy connects map -> regulate -> drain/stabilise -> warn -> evacuate -> update inventory . **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in "SUPPLEMENTARY CROSS-OWNED PYQ 9 - UPSC Mains GS-III 2021 Q18, 15 marks".

Executable exam-length answer / compression plan: For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

How to improve this answer: For "SUPPLEMENTARY CROSS-OWNED PYQ 9 - UPSC Mains GS-III 2021 Q18, 15 marks", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

SUPPLEMENTARY CROSS-OWNED PYQ 10 - UPSC Mains GS-I 2024 Q14, 15 marks

Exact question: The groundwater potential of the Gangetic valley is on a serious decline. How may it affect the food security of India? (Answer in 250 words)

Model answer

The Gangetic valley's alluvial aquifers support a major irrigated foodgrain region. Persistent withdrawal beyond replenishment can therefore transmit hydrological stress into food insecurity.

```
falling head/yield -> deeper pumping + higher energy cost
                   -> delayed/less reliable irrigation
                   -> crop-yield and income risk
                   -> food availability, access and stability stress
```

Small and marginal farmers have less capacity to deepen wells, purchase pumps or absorb energy costs, so the first effect is often unequal access to irrigation. Rice-wheat and other water-intensive rotations can face timing and yield risk, while reduced dry-season buffer increases dependence on variable rainfall. Higher production costs can affect prices and procurement; quality deterioration can also make nominal groundwater unusable.

Yet tube-well irrigation has supported productivity and drought resilience. The solution is not indiscriminate pumping prohibition. It is conjunctive surface-groundwater use, crop diversification aligned with agro-climate, efficient irrigation, rational energy/procurement signals, protection of recharge zones and wetlands, managed recharge where suitable, and aquifer-level community water budgets.

Conclusion: food security requires managing groundwater as production infrastructure and a common-pool resource, with equity for smallholders.

Why this earns marks: It links groundwater to all food-security pillars, explains the transmission chain and gives a balanced demand-side solution.

Demand decoding: The directive **answer** requires a direct position on "SUPPLEMENTARY CROSS-OWNED PYQ 10 - UPSC Mains GS-I 2024 Q14, 15 marks", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in "SUPPLEMENTARY CROSS-OWNED PYQ 10 - UPSC Mains GS-I 2024 Q14, 15 marks".

Analytical body:

1. **Claim and named evidence:** SUPPLEMENTARY CROSS-OWNED PYQ 10 - UPSC Mains GS-I 2024 Q14, 15 marks **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** Exact question: The groundwater potential of the Gangetic valley is on a serious decline. How may it affect the food security of India? (Answer in 250 words) **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** The Gangetic valley's alluvial aquifers support a major irrigated foodgrain region. Persistent withdrawal beyond replenishment can therefore transmit hydrological stress into food insecurity. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
4. **Claim and named evidence:** falling head/yield -> deeper pumping + higher energy cost **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
5. **Claim and named evidence:** food availability, access and stability stress **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in "SUPPLEMENTARY CROSS-OWNED PYQ 10 - UPSC Mains GS-I 2024 Q14, 15 marks".

Executable exam-length answer / compression plan: For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

How to improve this answer: For "SUPPLEMENTARY CROSS-OWNED PYQ 10 - UPSC Mains GS-I 2024 Q14, 15 marks", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

SUPPLEMENTARY CROSS-OWNED PYQ 11 - UPSC Mains GS-III 2025 Q8, 10 marks

Exact question: Seawater intrusion in the coastal aquifers is a major concern in India. What are the causes of seawater intrusion and the remedial measures to combat this hazard? (Answer in 150 words)

Model answer

Seawater intrusion is landward or upward movement of saline water into a coastal freshwater aquifer when freshwater hydraulic head becomes inadequate.

Causes: excessive pumping and clustered deep wells; reduced recharge through surface sealing or drought; coastal-aquifer geometry and preferential pathways; irrigation return/saline canals; and sea-level or storm stress where locally relevant. A pumping well may also draw saline water upward as **upconing**.

Measures: monitor groundwater head and salinity; cap, meter or redistribute abstraction; protect recharge zones; use suitably treated managed recharge; space and redesign wells; adopt efficient irrigation and less water-intensive uses; use site-specific subsurface/hydraulic barriers where feasible; and prevent pollution that compounds salinity.

Desalination may supply water but does not by itself restore aquifer pressure. Recharge without source-water and aquifer safeguards can worsen quality.

Conclusion: coastal aquifer security needs a monitored balance of pumping, recharge and land-use suited to local hydrogeology.

Why this earns marks: It begins with the hydraulic mechanism and matches each remedy to a cause.

Demand decoding: The directive **answer** requires a direct position on "SUPPLEMENTARY CROSS-OWNED PYQ 11 - UPSC Mains GS-III 2025 Q8, 10 marks", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in "SUPPLEMENTARY CROSS-OWNED PYQ 11 - UPSC Mains GS-III 2025 Q8, 10 marks".

Analytical body:

1. **Claim and named evidence:** SUPPLEMENTARY CROSS-OWNED PYQ 11 - UPSC Mains GS-III 2025 Q8, 10 marks **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** Exact question: Seawater intrusion in the coastal aquifers is a major concern in India. What are the causes of seawater intrusion and the remedial measures to combat this hazard? (Answer in 150 words) **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** Seawater intrusion is landward or upward movement of saline water into a coastal freshwater aquifer when freshwater hydraulic head becomes inadequate. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the

directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

4. **Claim and named evidence:** Desalination may supply water but does not by itself restore aquifer pressure. Recharge without source-water and aquifer safeguards can worsen quality. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
5. **Claim and named evidence:** Conclusion: coastal aquifer security needs a monitored balance of pumping, recharge and land-use suited to local hydrogeology. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in "SUPPLEMENTARY CROSS-OWNED PYQ 11 - UPSC Mains GS-III 2025 Q8, 10 marks".

Executable exam-length answer / compression plan: For a 10-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise three process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

How to improve this answer: For "SUPPLEMENTARY CROSS-OWNED PYQ 11 - UPSC Mains GS-III 2025 Q8, 10 marks", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

SUPPLEMENTARY CROSS-OWNED PYQ 12 - UPSC Mains GS-III 2025 Q13, 15 marks

Exact question: Examine the factors responsible for depleting groundwater in India. What are the steps taken by the government to mitigate such depletion of groundwater? (Answer in 250 words)

Model answer

Groundwater depletes when withdrawal persistently exceeds recharge at the relevant aquifer scale.

Factors: irrigation-dominated extraction; water-intensive crops and procurement incentives; low-cost/unmetered farm energy; numerous private wells; urban and industrial demand; leakage and sealed recharge surfaces; drought and rainfall variability; loss of tanks/wetlands; and limited storage in weathered/fractured hard-rock aquifers. Weak aquifer-scale regulation turns an individually rational pump into a collective decline.

Government steps: CGWB conducts annual dynamic resource assessment, level/quality monitoring and technical studies. NAQUIM 1.0 mapped about 25 lakh sq km; NAQUIM 2.0 targets detailed priority-area studies. The Atal Bhujal pilot used community water-security planning in 8,203 Gram Panchayats across seven states. JSA: Catch the Rain, JSJB: CTR 2026, the Master Plan for Artificial Recharge, PMKSY components, water-body restoration and Jal Shakti Kendras support recharge and conservation. CGWA/state rules regulate specified extraction.

The 6-10 August 2026 Ministry replies report national recharge/extraction improvement and large structure counts, but these are not proof of recovery in every aquifer. Crop/energy incentives, metering, reuse, maintenance, quality safeguards and transparent local outcomes remain necessary.

Conclusion: scientific mapping and recharge must be coupled with demand reform and accountable state/local governance.

Why this earns marks: It examines natural, economic and institutional drivers and evaluates government measures rather than listing schemes.

Demand decoding: The directive **answer** requires a direct position on "SUPPLEMENTARY CROSS-OWNED PYQ 12 - UPSC Mains GS-III 2025 Q13, 15 marks", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in "SUPPLEMENTARY CROSS-OWNED PYQ 12 - UPSC Mains GS-III 2025 Q13, 15 marks".

Analytical body:

1. **Claim and named evidence:** SUPPLEMENTARY CROSS-OWNED PYQ 12 - UPSC Mains GS-III 2025 Q13, 15 marks **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** Exact question: Examine the factors responsible for depleting groundwater in India. What are the steps taken by the government to mitigate such depletion of groundwater? (Answer in 250 words) **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** Groundwater depletes when withdrawal persistently exceeds recharge at the relevant aquifer scale. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
4. **Claim and named evidence:** Conclusion: scientific mapping and recharge must be coupled with demand reform and accountable state/local governance. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
5. **Claim and named evidence:** Why this earns marks: It examines natural, economic and institutional drivers and evaluates government measures rather than listing schemes. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in "SUPPLEMENTARY CROSS-OWNED PYQ 12 - UPSC Mains GS-III 2025 Q13, 15 marks".

Executable exam-length answer / compression plan: For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

How to improve this answer: For "SUPPLEMENTARY CROSS-OWNED PYQ 12 - UPSC Mains GS-III 2025 Q13, 15 marks", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

ORIGINAL 10-MARK QUESTION 1 - process distinction

Question: Distinguish weathering, erosion and mass movement. Show how their interaction can create risk along a Himalayan road corridor. (150 words)

Model answer: Weathering breaks or alters rock in situ; erosion detaches and transports material through water, wind, ice or waves; mass movement transfers rock/debris downslope mainly under gravity. In a Himalayan road corridor, fractured rock and weathered regolith provide susceptible material. A road cut may steepen or undercut the slope and redirect drainage. Intense rain can infiltrate joints, add weight and raise pore-water pressure until resisting strength is exceeded. The resulting slide blocks the road; runoff may then entrain debris and erode the toe. GSI inventory/susceptibility work shows why geology, slope, rainfall and exposure must be combined. Yet neither road construction nor rainfall alone proves cause at a particular site. Site investigation, drainage design, toe support, slope treatment, spoil control, maintenance and warning/closure protocols are needed. **Why this earns marks:** definitions are exact, the causal chain is applied to India, and attribution is qualified.

Demand decoding: The directive **answer** requires a direct position on "Distinguish weathering, erosion and mass movement. Show how their interaction can create risk...", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in "Distinguish weathering, erosion and mass movement. Show how their interaction can create risk along a Himalayan road corridor. (150 words)".

Analytical body:

1. **Claim and named evidence:** ORIGINAL 10-MARK QUESTION 1 - process distinction **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in "Distinguish weathering, erosion and mass movement. Show how their interaction can create risk along a Himalayan road corridor. (150 words)".

Executable exam-length answer / compression plan: For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

How to improve this answer: For "Distinguish weathering, erosion and mass movement. Show how their interaction can create risk...", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

ORIGINAL 10-MARK QUESTION 2 - porosity and permeability

Question: Explain why porosity alone cannot determine whether a rock or sediment is a productive aquifer. (150 words)

Model answer: Porosity measures the fraction of void space, whereas permeability measures whether pores or fractures are connected well enough to transmit water. Clay can contain abundant microscopic pore space yet release and transmit water very slowly; it can act as an aquitard. Well-sorted sand and gravel combine useful porosity with connected pores. Chalk may absorb and transmit water through matrix pores and fractures. Crystalline hard rock has little primary porosity but can yield groundwater through weathered mantles, joints and fractures. Productivity also depends on saturated thickness, recharge, boundary conditions and well design. Therefore aquifer assessment requires both storage and transmissivity evidence. **Why this earns marks:** it uses the exact close-option distinction, gives contrasting materials and connects the concept to India's hard-rock aquifers.

Demand decoding: The directive **explain** requires a direct position on "Explain why porosity alone cannot determine whether a rock or sediment is a productive...", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in "Explain why porosity alone cannot determine whether a rock or sediment is a productive aquifer. (150 words)".

Analytical body:

1. **Claim and named evidence:** ORIGINAL 10-MARK QUESTION 2 - porosity and permeability **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in "Explain why porosity alone cannot determine whether a rock or sediment is a productive aquifer. (150 words)".

Executable exam-length answer / compression plan: For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim ->

named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

How to improve this answer: For "Explain why porosity alone cannot determine whether a rock or sediment is a productive...", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

ORIGINAL 15-MARK QUESTION 1 - watershed management

Question: Examine why soil erosion and slope failure should be managed at watershed scale in India. Illustrate with distinct physiographic settings. (250 words)

Model answer: Soil erosion and slope failure are linked watershed processes because vegetation, slope length, drainage and infiltration control runoff energy, sediment transfer and pore-water conditions. In the Chambal tract, diffuse sheet wash can concentrate into rills, gullies and ravines; cover, contouring, gully-head control and catchment drainage must act upslope. Himalayan corridors combine high relief, fractured/weathered material, river undercutting, road cuts and intense rainfall; drainage, toe protection, designed stabilisation and hazard-sensitive alignment must work together. Western Ghats/Nilgiris slopes have a different mix of deep weathering/laterite, escarpment relief, orographic rain, quarry/road cuts and altered drainage. Catchment treatment can also improve infiltration and spring/aquifer recharge, but poorly sited sediment-laden structures may clog or destabilise slopes. Watershed scale does not eliminate plot-level tenure, maintenance and incentive problems. India therefore needs nested plot-slope-micro-watershed-aquifer planning supported by monitoring and community institutions. **Why this earns marks:** it examines a causal proposition, compares regions and matches interventions to mechanisms.

Demand decoding: The directive **examine** requires a direct position on "Examine why soil erosion and slope failure should be managed at watershed scale in India....", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in "Examine why soil erosion and slope failure should be managed at watershed scale in India. Illustrate with distinct physiographic settings. (250 words)".

Analytical body:

1. **Claim and named evidence:** ORIGINAL 15-MARK QUESTION 1 - watershed management **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in "Examine why soil erosion and slope failure should be managed at watershed scale in India. Illustrate with distinct physiographic settings. (250 words)".

Executable exam-length answer / compression plan: For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

How to improve this answer: For "Examine why soil erosion and slope failure should be managed at watershed scale in India....", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

ORIGINAL 15-MARK QUESTION 2 - urban subsidence

Question: Analyse the relationship between groundwater extraction, geological conditions and land subsidence in Indian cities and plains. Suggest a risk-reduction framework. (250 words)

Model answer: Groundwater extraction can cause land subsidence where pumping lowers pore pressure in compressible aquifer-system sediments, transfers load to the grain skeleton and produces compaction. The effect is strongest where thick fine-grained layers are susceptible, withdrawal is concentrated and recharge is inadequate. Hard-rock cities may show a different response because storage is fracture-controlled; not every falling water table produces measurable subsidence. Urban loading, excavation, drainage leakage, cavities and natural consolidation can also contribute, so attribution requires geodetic, borehole, pumping and hydrogeological evidence. Impacts include differential settlement, cracked buildings, damaged drains and increased flood exposure. Risk reduction requires aquifer-wise extraction budgets; well registration and metering for major users; conjunctive supply and reuse; recharge-zone protection and safe managed recharge; InSAR/GNSS and groundwater-level monitoring; geotechnical zoning; building/drainage audits; and public disclosure. **Conclusion:** subsidence policy must connect water management with geotechnical land-use planning while avoiding single-cause claims. **Why this earns marks:** it explains the compaction mechanism, distinguishes settings and proposes measurable interventions.

Demand decoding: The directive **analyse** requires a direct position on "Analyse the relationship between groundwater extraction, geological conditions and land...", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in "Analyse the relationship between groundwater extraction, geological conditions and land subsidence in Indian cities and plains. Suggest a risk-...".

Analytical body:

1. **Claim and named evidence:** ORIGINAL 15-MARK QUESTION 2 - urban subsidence **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in "Analyse the relationship between groundwater extraction, geological conditions and land subsidence in Indian cities and plains. Suggest a risk-...".

Executable exam-length answer / compression plan: For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

How to improve this answer: For "Analyse the relationship between groundwater extraction, geological conditions and land...", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

ORIGINAL 20-MARK QUESTION 1 - integrated landslide governance

Question: "A landslide disaster is produced as much by exposure and vulnerability as by slope failure." Analyse for India and propose an integrated strategy. (250 words)

Model answer: A landslide hazard is the physical probability and intensity of slope movement; disaster risk emerges when people and assets are exposed and vulnerable, while capacity is weak. India's Himalayan and North-Eastern belts combine high relief, fractured or weathered materials, seismicity, river incision and monsoon triggers. Western Ghats/Nilgiris commonly add deep weathering, orographic rain, slope cutting, quarrying and drainage disruption. Yet the same physical event produces different losses depending on settlement location, road redundancy, construction quality, poverty and warning access. GSI inventory and susceptibility mapping help avoid high-risk siting; the June 2026 Chanmari report shows why field diagnosis of joints, toe erosion, runoff and settlement drainage is still necessary. An integrated strategy should: regulate land use and carrying capacity; require site investigation and safe road/slope design; maintain catch drains and culverts; combine engineered stabilisation with bioengineering; monitor rainfall and movement; link forecasts to local communication, closures and evacuation; train communities; protect alternate access; and update inventories after events. Forecasts cannot stabilise slopes, and retaining walls cannot compensate for unmanaged water. **Conclusion:** India must join source treatment with exposure reduction and social preparedness. **Why this earns marks:** it uses the risk equation, regional differentiation, current GSI evidence and a layered strategy.

Demand decoding: The directive **analyse** requires a direct position on ""A landslide disaster is produced as much by exposure and vulnerability as by slope failure."...", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in ""A landslide disaster is produced as much by exposure and vulnerability as by slope failure." Analyse for India and propose an integrated strategy....".

Analytical body:

1. **Claim and named evidence:** ORIGINAL 20-MARK QUESTION 1 - integrated landslide governance **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in ""A landslide disaster is produced as much by exposure and vulnerability as by slope failure." Analyse for India and propose an integrated strategy....".

Executable exam-length answer / compression plan: For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

How to improve this answer: For ""A landslide disaster is produced as much by exposure and vulnerability as by slope failure."...", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

ORIGINAL 20-MARK QUESTION 2 - groundwater common-pool resource

Question: "Groundwater is a stock governed by flows, quality and institutions-not an invisible reserve waiting to be pumped." Analyse this proposition for India, including food security and coastal aquifers. (250 words)

Model answer: Groundwater storage changes through recharge, inflow, natural discharge, baseflow, pumping and inter-aquifer leakage; usable supply also depends on permeability and quality. The 2025 assessment reported in August 2026 distinguishes 448.52 BCM recharge, 407.75 BCM extractable resource and 247.22 BCM extraction. These national values cannot certify a local aquifer. In the Gangetic food belt, falling head raises lift and energy costs, reduces irrigation reliability and affects output, smallholder income and food-security stability. In hard-rock India, limited discontinuous weathered/fracture storage creates sharp local variability. At the coast, excessive pumping lowers freshwater head, enabling landward saline-interface movement and upconing. Arsenic, fluoride, nitrate and salinity further reduce usable stock through different pathways. CGWB assessment/monitoring, NAQUIM, Atal Bhujal learning and recharge programmes provide tools, but sustainable management also needs crop-energy reform, efficient irrigation, conjunctive use, reuse, quality safeguards, state/local regulation and aquifer-level participation. **Conclusion:** recharge is necessary but cannot substitute for demand management and accountable institutions. **Why this earns marks:** it integrates stock-flow, regional hydrogeology, quality, food and governance with current evidence.

DIAGRAM AND MAP PRACTICE

Diagram 1 - weathering to storage

```
ROCK -> weathering -> regolith -> [gravity movement / agent transport]
                                  -> deposition/storage -> remobilisation
```

Diagram 2 - slope force and water

```
rain -> infiltration -> weight + pore pressure -> lower effective strength
road/river cut -> toe support loss --------------------^ -> failure
```

Diagram 3 - aquifer section

```
recharge area -> unconfined water table -> aquitard -> confined aquifer
                                        well rises to potentiometric level
```

Diagram 4 - coastal upconing

```
coast | freshwater wedge over saline water
      | pumping well -> cone of depression -> saline rise/upconing
```

Map 5 - India recall

```
Himalaya/NE: landslides + springs | Western Ghats/Nilgiris: rain-weathered slopes
Chambal: ravines | Indo-Gangetic plain: alluvial aquifers/food
Peninsula: hard-rock aquifers | Coast: saline interface
```

Answer skeleton 6

```
definition -> labelled diagram/map -> mechanism -> India evidence
-> effect/evaluation -> limitation -> matched response -> qualified conclusion
```

Demand decoding: The directive **analyse** requires a direct position on ""Groundwater is a stock governed by flows, quality and institutions-not an invisible reserve...", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in ""Groundwater is a stock governed by flows, quality and institutions-not an invisible reserve waiting to be pumped." Analyse this proposition for...".

Analytical body:

1. **Claim and named evidence:** ORIGINAL 20-MARK QUESTION 2 - groundwater common-pool resource **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** ROCK -> weathering -> regolith -> [gravity movement / agent transport] **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** rain -> infiltration -> weight + pore pressure -> lower effective strength **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
4. **Claim and named evidence:** road/river cut -> toe support loss --------------------^ -> failure **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
5. **Claim and named evidence:** recharge area -> unconfined water table -> aquitard -> confined aquifer **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date

boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in ""Groundwater is a stock governed by flows, quality and institutions-not an invisible reserve waiting to be pumped." Analyse this proposition for...".

Executable exam-length answer / compression plan: For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

How to improve this answer: For ""Groundwater is a stock governed by flows, quality and institutions-not an invisible reserve...", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.